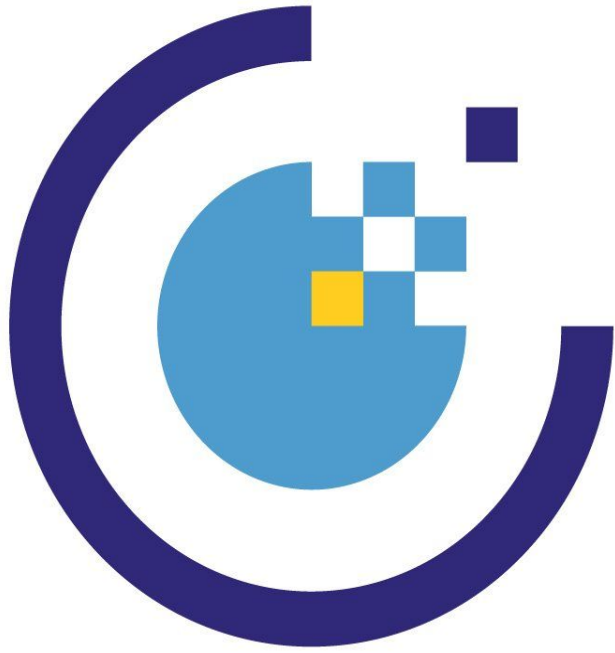




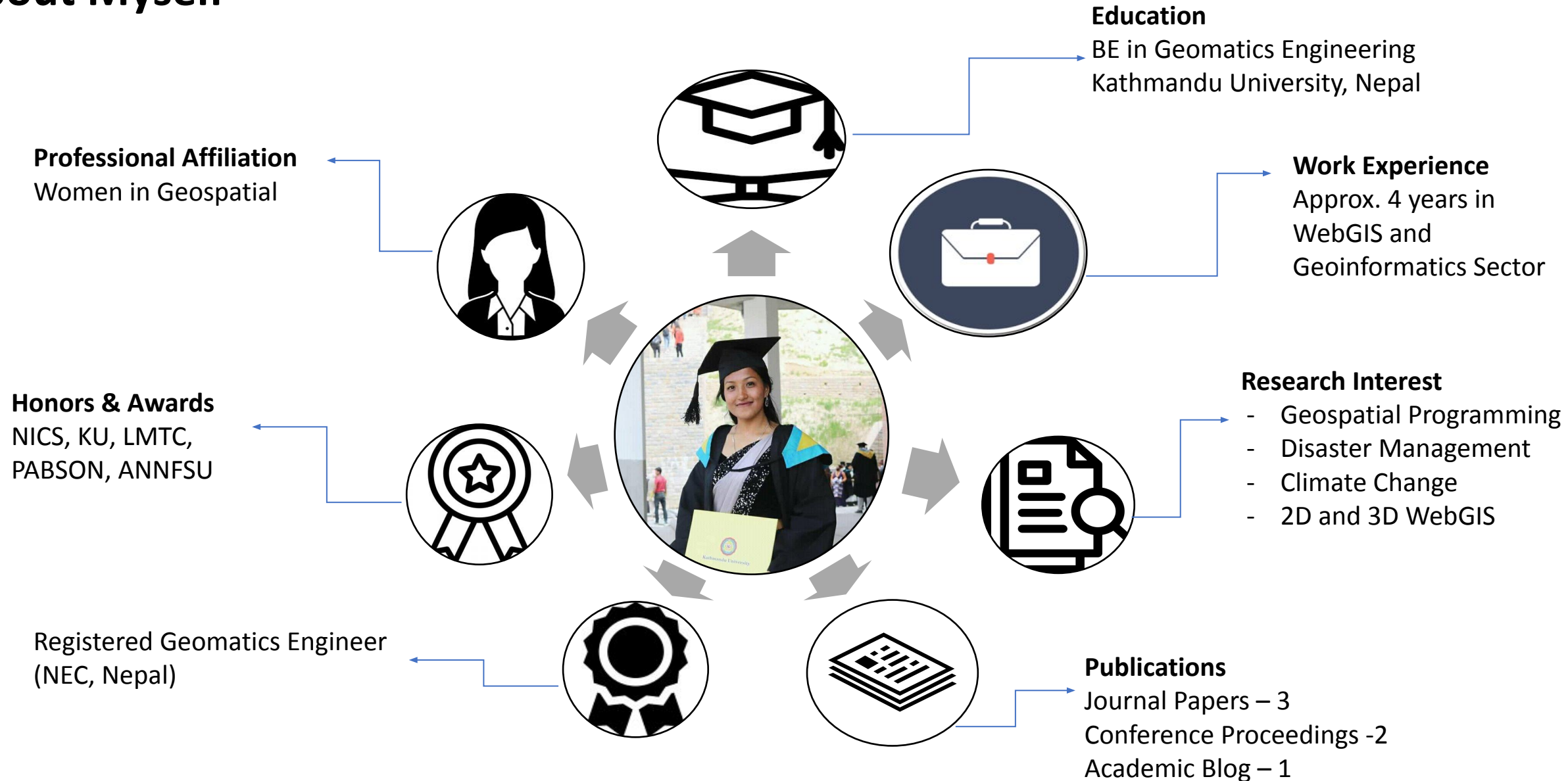
**Copernicus Master in  
Digital Earth  
Specialization  
in GeoData Science**



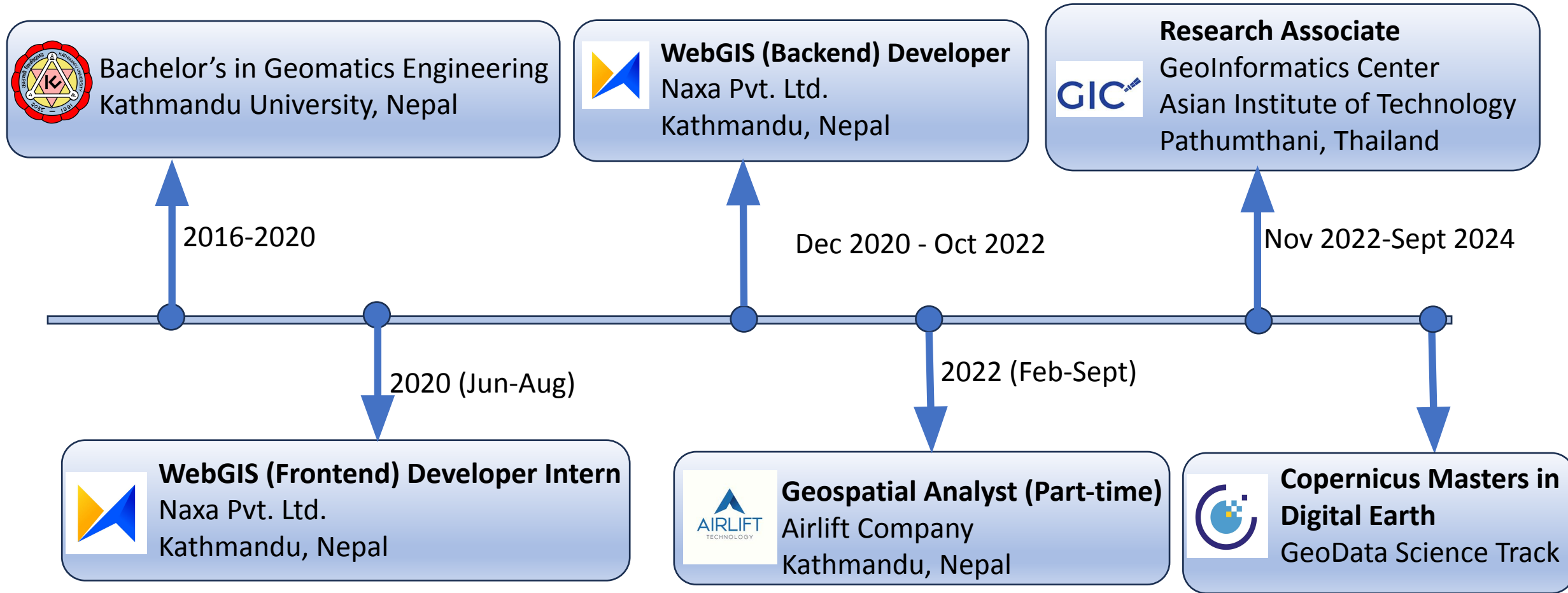
**Rabina Twayana**

Email: [rabina.twayana@stud.plus.ac.th](mailto:rabina.twayana@stud.plus.ac.th)  
ePortfolio: <http://rabinatwayana.com.np>

# About Myself

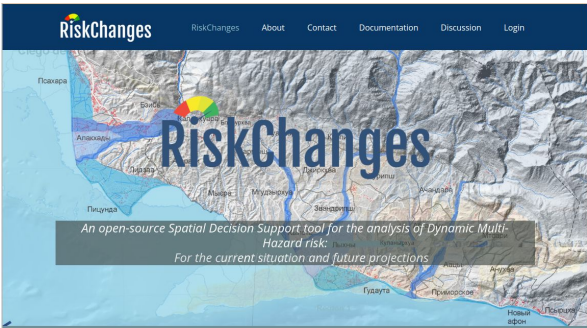


# Academic and Professional Journey

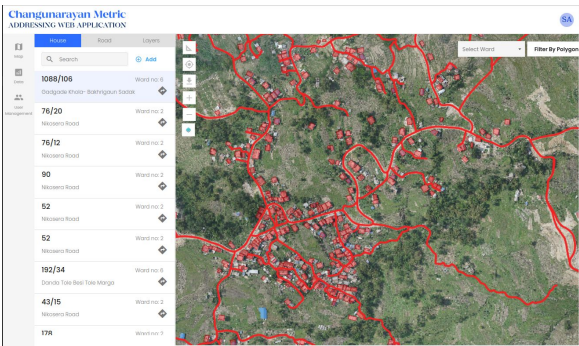




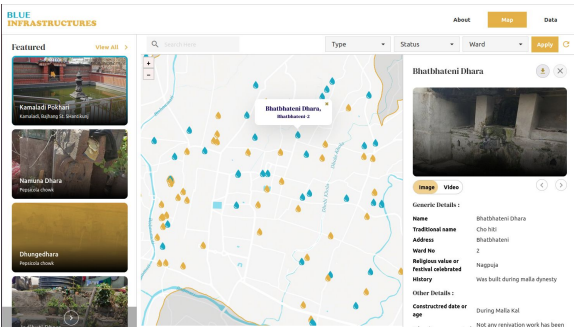
# Projects



RiskChanges



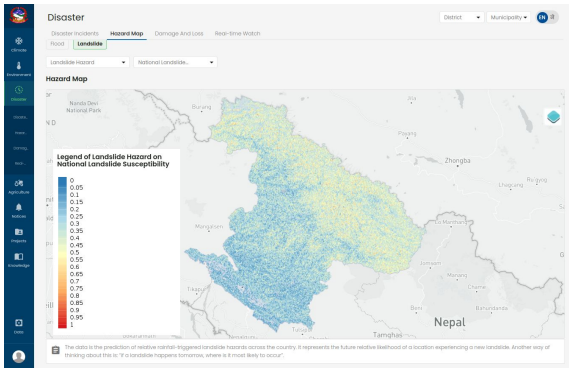
Changu Metric Addressing System



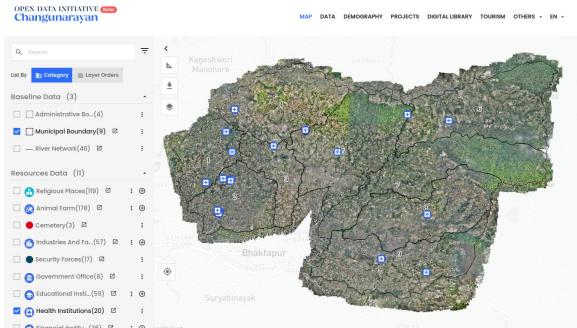
Blue Infrastructure Mapping



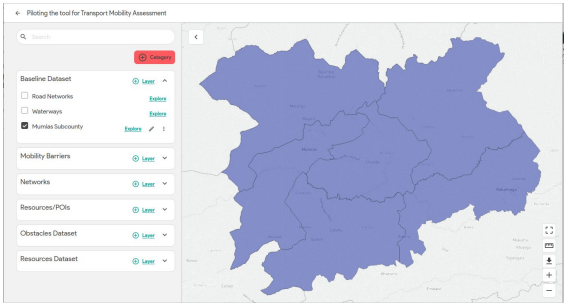
Cleanup Nepal



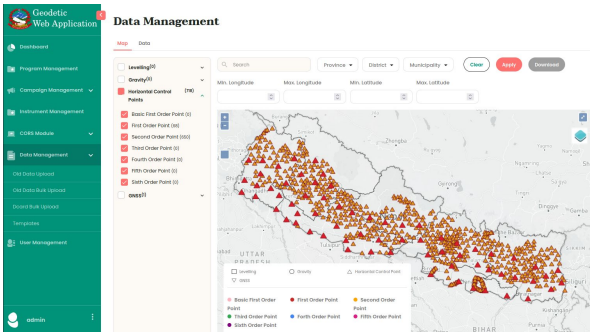
Provincial Climate Change Management Information System



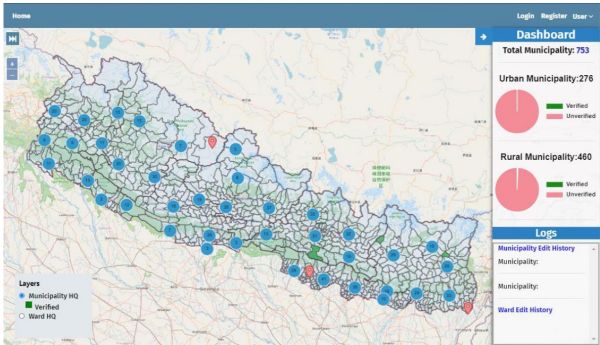
Changunarayan Open data Portal



Participatory GIS Mapping



Geodetic Web Application



Municipal Headquarter Mapping Initiative

# Publications



Rabina Twayana

Geomatics Engineer, Kathmandu University  
Verified email at ait.asia  
WebGIS Remote Sensing GIS

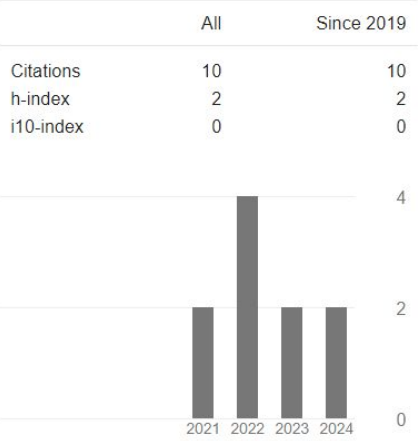
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| TITLE                                                                                                                                                                                                                                 | CITED BY | YEAR |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|
| Future land use land cover scenario simulation using open-source GIS for the City of Banepa and Dhulikhel municipality, Nepal<br>S Bhandari, R Twayana, R Shrestha, K Sharma<br>FOSS 4G-ASIA                                          | 5        | 2021 |
| Analyzing Urban Growth Pattern and Driving Factors Using Remote Sensing and GIS: A Case Study of Banepa Municipality, Nepal<br>R Twayana, S Bhandari, R Shrestha<br>Journal on Geoinformatics, Nepal 20 (1), 9-18                     | 4        | 2021 |
| MONITORING LAND COVER/LAND USE CHANGE USING REMOTE SENSING AND GIS TECHNIQUE AT DHULIKHEL AND BANEPA MUNICIPALITY, NEPAL<br>R Shrestha, S Bhandari, R Twayana<br>International Journal of Current Research in Science Engineering ... | 1        | 2021 |
| CHANGE DETECTION OF BUILT UP AREAS APPLYING BUILT-UP INDEX FOR KATHMANDU DISTRICT<br>S Baidar, S Bhandari, A Joshi, R Twayana                                                                                                         |          |      |
| LAND TENURE SYSTEM IN ETHIOPIA: ISSUES, CHALLENGES AND OPPORTUNITIES<br>R Twayana, B Baniya, S Kuikel, R Saud                                                                                                                         |          |      |
| PROBLEMS AND ITS POSSIBLE SOLUTIONS FOR CONVERSION OF 2D CADASTRE TO 3D CADASTRE IN CONTEXT OF NEPAL<br>R Twayana, S Bhandari                                                                                                         |          |      |

Articles 1–6 SHOW MORE

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# Conference Presentation

## **RiskChanges: A WebGIS Application for Multi-Hazard Quantitative Risk Assessment**

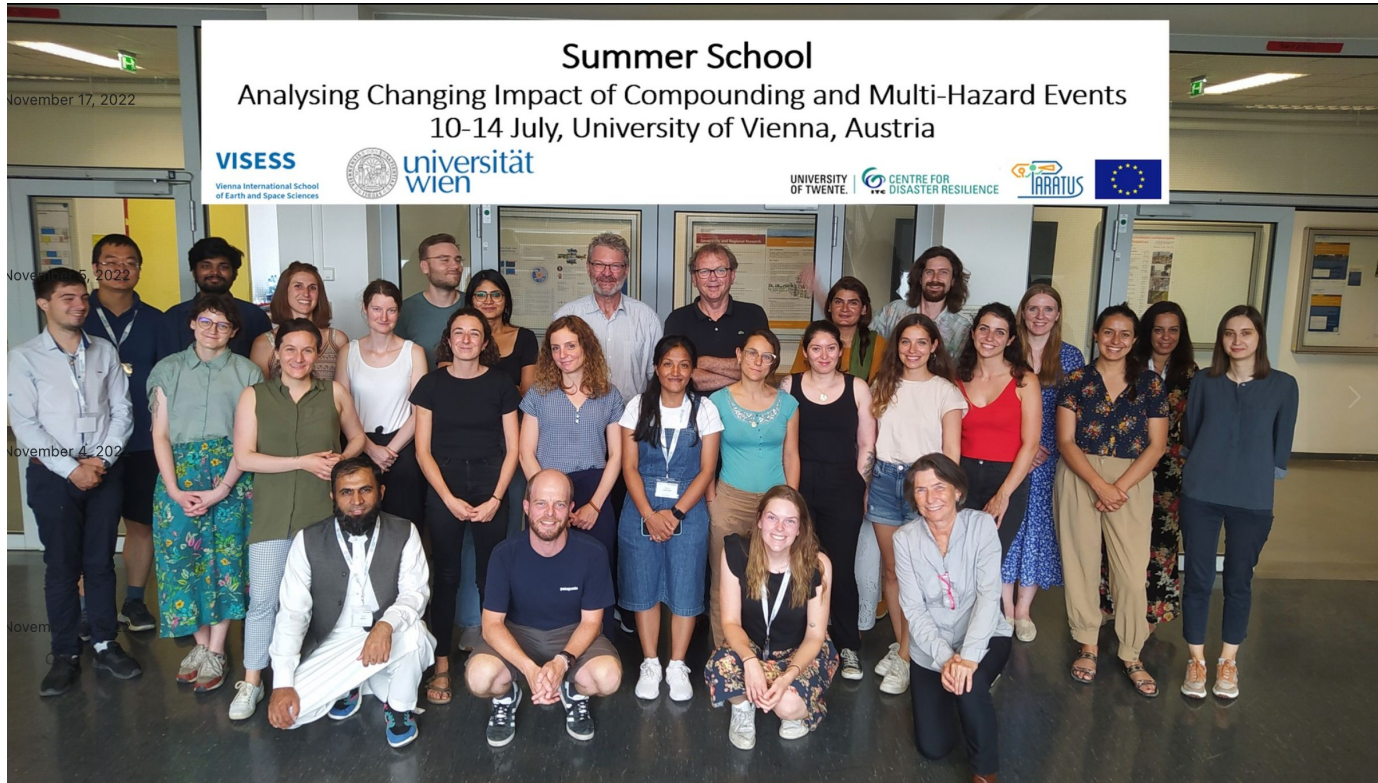
**Rabina Twayana**<sup>1\*</sup>, Cees Van Westen<sup>2</sup>, Manzul Kumar Hazarika<sup>3</sup>,  
Ashok Dahal<sup>4</sup>, Anish Ratna Shakya<sup>5</sup>

<sup>1</sup>Asian Institute of Technology (AIT), <sup>2</sup>University of Twente (UT), <sup>3</sup>Asian Institute of Technology (AIT),  
<sup>4</sup>University of Twente (UT), <sup>5</sup>Asian Institute of Technology (AIT)

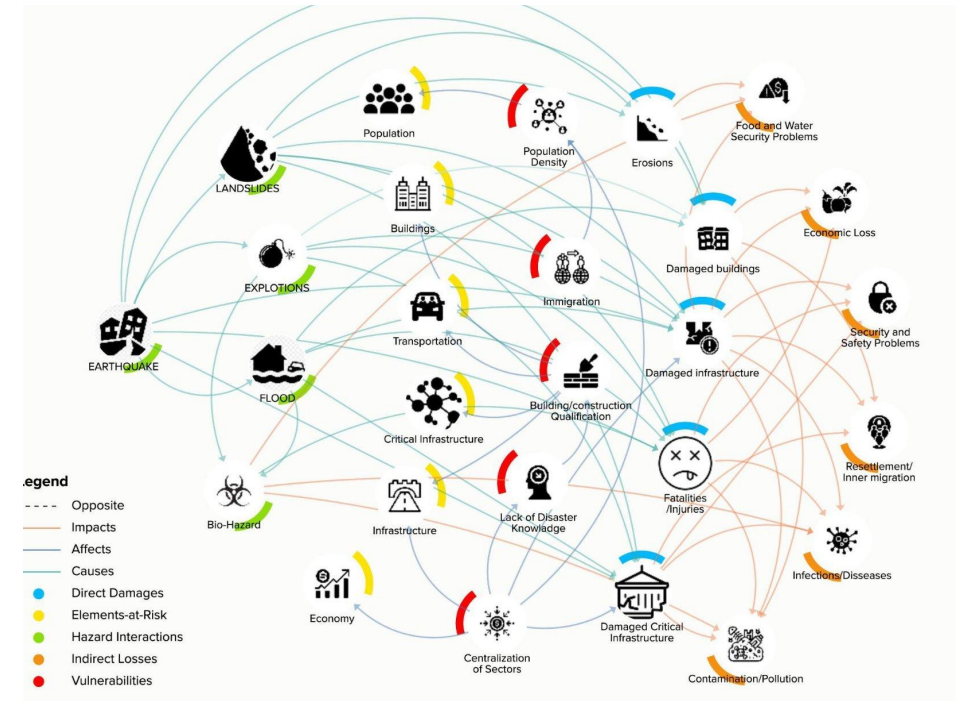
\*Corresponding email : [rabina@ait.asia](mailto:rabina@ait.asia)



# Summer School (2023)



## Impact chains of a large earthquake in Istanbul in Urban Sector



**Aim:** To analyze the complex impact chains of hazard interactions, exposure, vulnerability, direct and indirect impacts within different sectors in one of five case study areas: Alps, Turkey, Romania, Caribbean and Pakistan



# Outreach

UNDRR

PreventionWeb

[Home](#) [Understanding disaster risk](#) [Knowledge Base](#) [Community](#) [Sendai Framework](#) 

[Home](#) > [drr community voices](#)

17 October 2023

## An open-source tool to assist in multi-hazard risk assessment

 Upload your content

Author(s): Rabina Twayana



UNDRR / RiskChanges

<https://www.preventionweb.net/drr-community-voices/open-source-tool-assist-multi-hazard-risk-assessment>





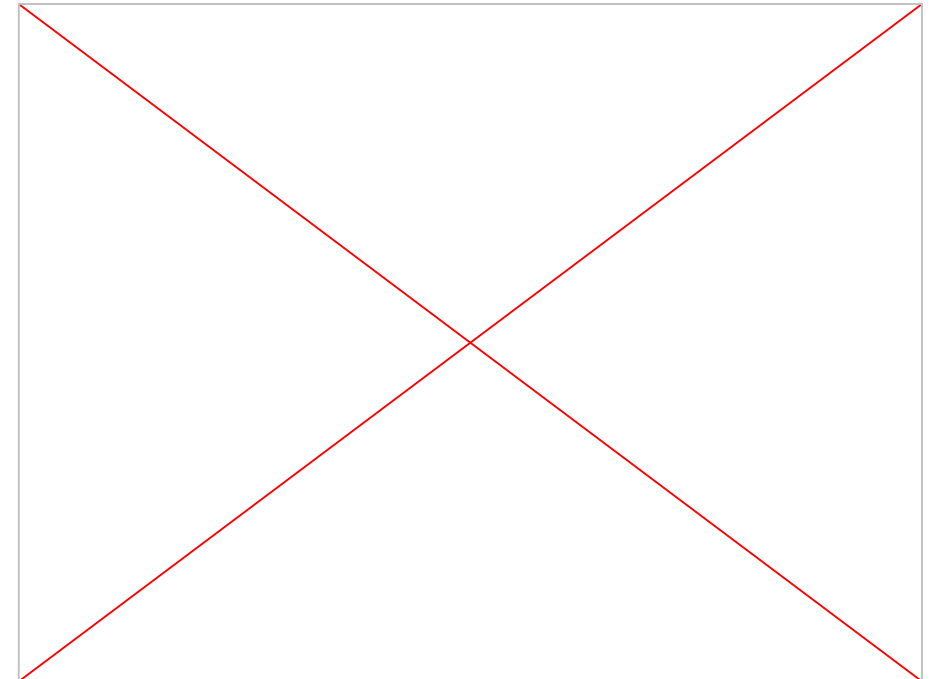
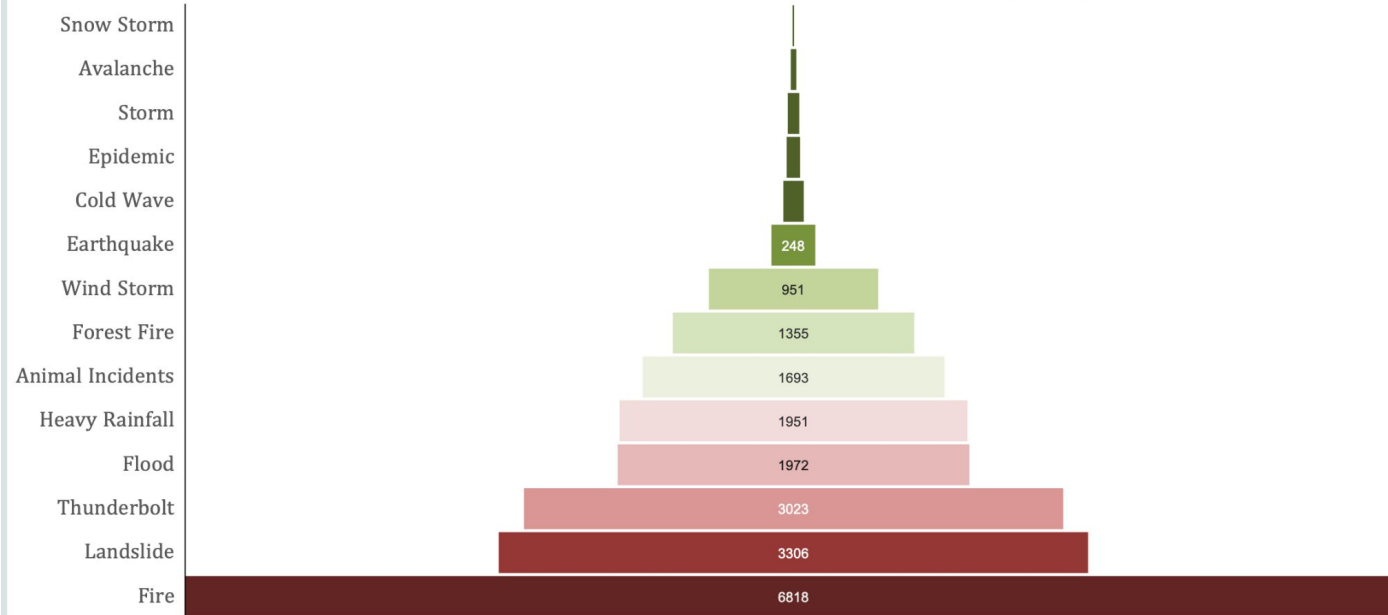
# Women in Geospatial Mentorship Programme

## Temporal Analysis of Hazard Occurrences in Nepal

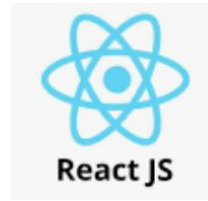
**Authors: Bibechana Sharma, Rabina Twayana, Allyssa Marie Arabit, Unnati Dhakal**

An analysis of temporal natural hazard occurrences in Nepal from 2011 to 2024 at the province level, depicted through a dynamic bar chart race and animated spatial map showcasing annual variations.

Total Natural Hazard Occurrence from 2011 to 2024 (April)



# Skills and Technologies



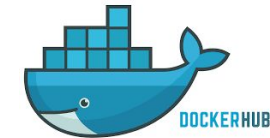
Google Cloud Platform  
**Cloud Computing**

**Geospatial Technologies**

**Programming**

**Database Management**

  
**Operating System**



**Containerization**

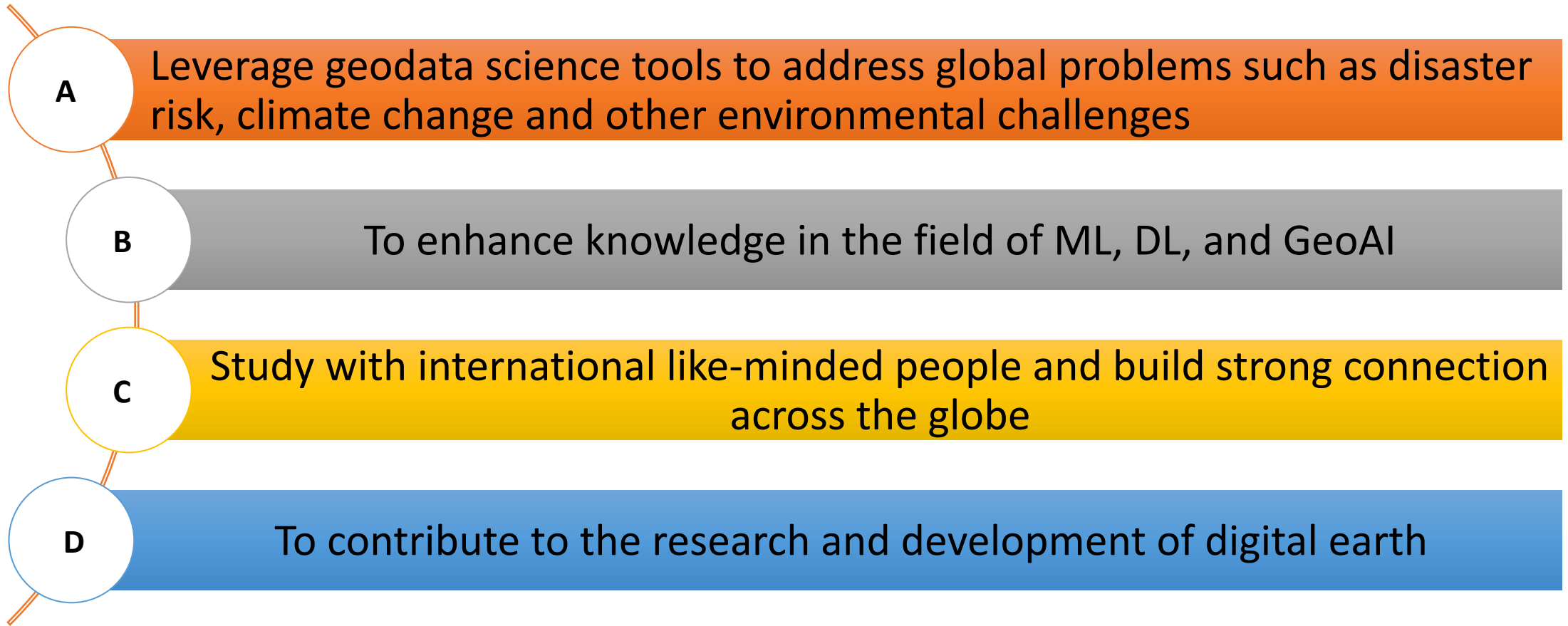


**Version Control**

**Geospatial Libraries and MapServer**

**Geospatial Software**

# Motivation





*Thank  
you*

